

Chapter 2

Reasoning and Proof

Unit 1 — Lines and Angles

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Key Vocabulary

Term	Definition
Inductive reasoning	Using patterns/examples to form a conjecture
Conjecture	A conclusion based on inductive reasoning
Counterexample	One example that disproves a conjecture
Conditional	An if-then statement: $p \rightarrow q$
Hypothesis	The "if" part (p) of a conditional
Conclusion	The "then" part (q) of a conditional
Deductive reasoning	Using accepted facts/laws to reach a conclusion

Conditional Statements

A **conditional** has the form: **If p , then q** ($p \rightarrow q$)

Form	Statement
Conditional	If p , then q
Converse	If q , then p
Inverse	If $\sim p$, then $\sim q$
Contrapositive	If $\sim q$, then $\sim p$

A conditional and its **contrapositive** are logically equivalent.
The converse and inverse are logically equivalent to each other.

Example — Conditional Forms

Conditional: If a figure is a square, then it has four sides.

Form	Statement
Converse	If a figure has four sides, then it is a square. <i>(False)</i>
Inverse	If a figure is not a square, then it doesn't have four sides. <i>(False)</i>
Contrapositive	If a figure doesn't have four sides, then it is not a square. <i>(True)</i>

Biconditional Statement

A **biconditional** is true when the conditional AND its converse are both true.

$$p \leftrightarrow q \quad \text{"}p \text{ if and only if } q\text{"}$$

Example: An angle is a right angle **if and only if** its measure is 90° .

Law of Detachment

If $p \rightarrow q$ is true and p is true, then q is true.

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Example:

- If it rains, the ground is wet. (*true*)
- It is raining. (*true*)
- $\therefore$ The ground is wet.

Law of Syllogism

If $p \rightarrow q$ and $q \rightarrow r$ are both true, then $p \rightarrow r$ is true.

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

Example:

- If a number ends in 0, it is divisible by 10.
- If a number is divisible by 10, it is divisible by 5.
- $\therefore$ If a number ends in 0, it is divisible by 5.

Postulate 2-1: Midpoint Uniqueness

A segment has **exactly one midpoint**.

Postulate 2-2: Angle Bisector Uniqueness

An angle has **exactly one bisector**.

Theorem 2-1: Properties Used in Proofs

Reflexive Property: $a = a$ (a quantity equals itself)

Symmetric Property: If $a = b$, then $b = a$

Transitive Property: If $a = b$ and $b = c$, then $a = c$

Substitution Property: If $a = b$, then a may replace b in any equation

Theorem 2-2: Segment Congruence Properties

Segment congruence is:

- **Reflexive:** $\overline{AB} \cong \overline{AB}$
- **Symmetric:** If $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$
- **Transitive:** If $\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$

Theorem 2-3: Angle Congruence Properties

Angle congruence is:

- **Reflexive:** $\angle A \cong \angle A$
- **Symmetric:** If $\angle A \cong \angle B$, then $\angle B \cong \angle A$
- **Transitive:** If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$

Theorem 2-4: Supplementary Angle Pairs

If two angles are supplementary to the **same angle** (or congruent angles), then they are congruent.

If $\angle 1$ and $\angle 2$ are supp., and $\angle 1$ and $\angle 3$ are supp., then $\angle 2 \cong \angle 3$

Theorem 2-5: Complementary Angle Pairs

If two angles are complementary to the **same angle** (or congruent angles), then they are congruent.

Theorem 2-6: Right Angles

All **right angles** are congruent.

$$\angle A = 90^\circ \text{ and } \angle B = 90^\circ \Rightarrow \angle A \cong \angle B$$

Theorem 2-7: Perpendicular Lines

If two lines are perpendicular, they form **four right angles**.

$$\ell \perp m \Rightarrow \text{all four angles} = 90^\circ$$

Chapter 2 — Theorem Summary

#	Name	Key Idea
—	Law of Detachment	$p \rightarrow q, p \Rightarrow q$
—	Law of Syllogism	$p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$
Thm 2-1	Algebraic Properties	Reflexive, Symmetric, Transitive, Substitution
Thm 2-2	Segment Congruence	Reflexive, symmetric, transitive
Thm 2-3	Angle Congruence	Reflexive, symmetric, transitive
Thm 2-4	Supp. to Same	Two angles supp. to same $\rightarrow$ congruent
Thm 2-5	Comp. to Same	Two angles comp. to same $\rightarrow$ congruent
Thm 2-6	Right Angles	All right angles are congruent
Thm 2-7	Perpendicular	$\perp$ lines form 4 right angles